

Software Design using Archetypes- Part #1

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Praseed Pai

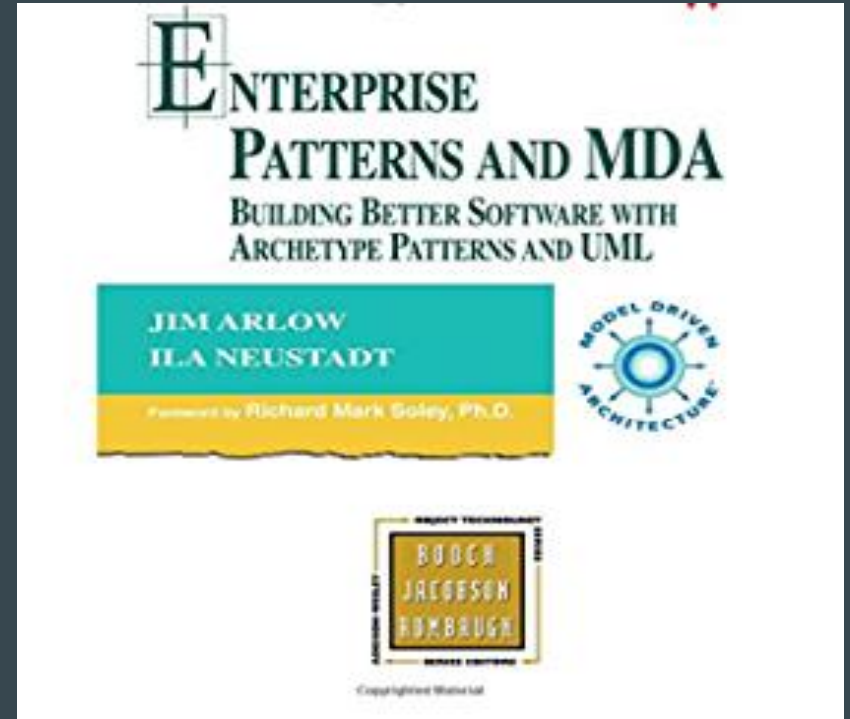
About the Presenter

- ◆ A Seasoned Software Engineering Professional with more than twenty five years of Exposure
- ◆ CO-Author of Two books on Computer Programming
- ◆ Explorer in “Philosophical Tools for Software Engineering” (Has Presented on it, Written one university accredited paper, Designed a Pattern based on Advaita Vedanta to transition from OOP to FRP)
- ◆ An Expert level professional in Cross Cultural Encounters (How to deal with a Russian/Eastern European? , Working with Racial stereotypes like Jews / Chinese / Latin Americans)
- ◆ A Critique of Digital Technology Fads (Programmers will be better off , if they stick to Programming. Do not run after so called AI/ML, BlockChain etc) - “Plumbing is preferred over Painting!”
- ◆ I also help Programmers eliminate their “Math-Phobia”
- ◆ Currently, EVP (Platform Engineering) for ChistaDATA inc.



INSPIRATION

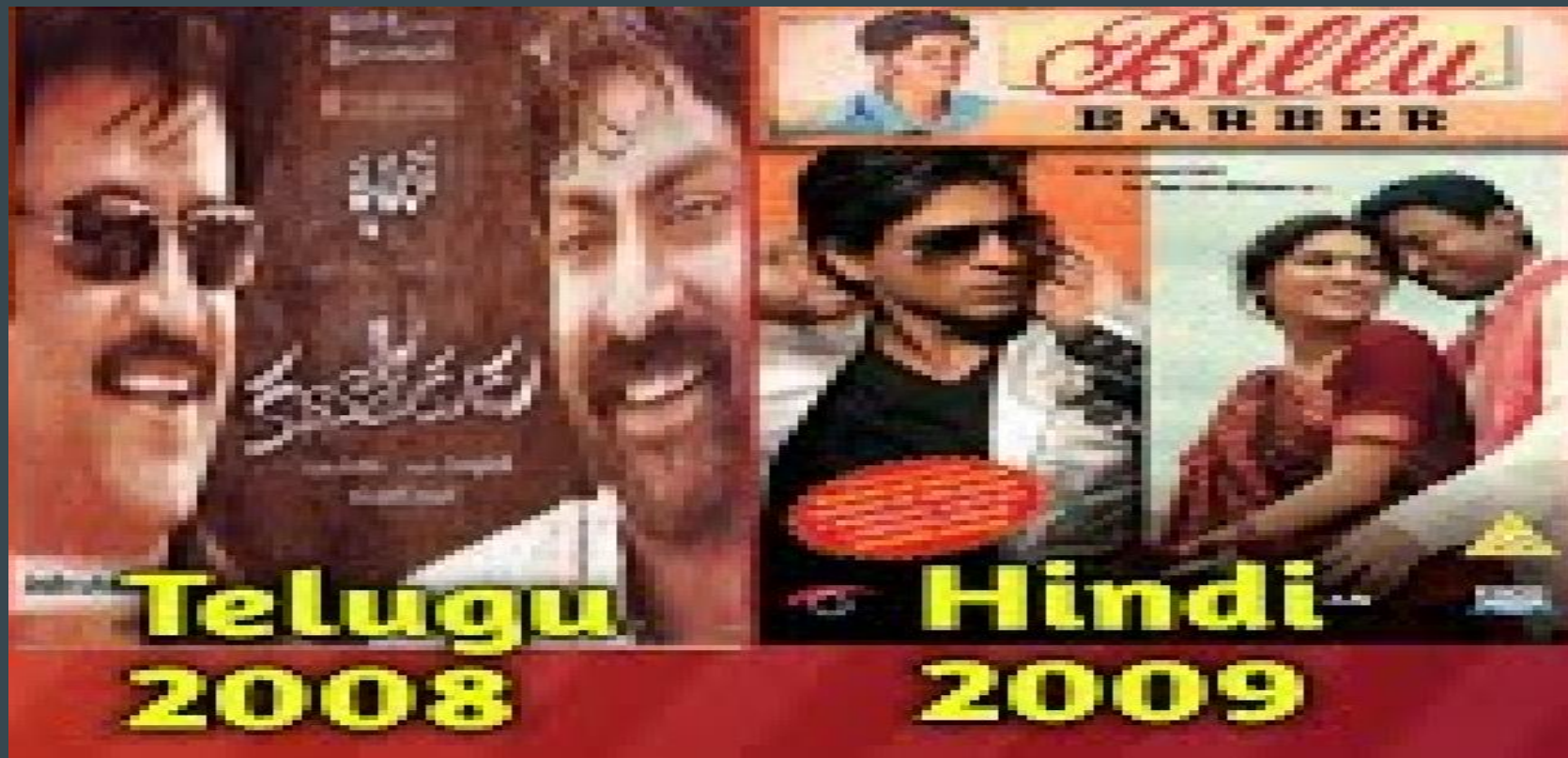
ENTERPRISE PATTERNS AND
MDA by Jim Arlow/Ila Neustadt
- Building Better Software with
Archetype Patterns and UML



A Successful Movie



The Film was successful in Telugu and Hindi too



The Real Reason for the Success of Movie (2008)



Run Away Success of HSSH



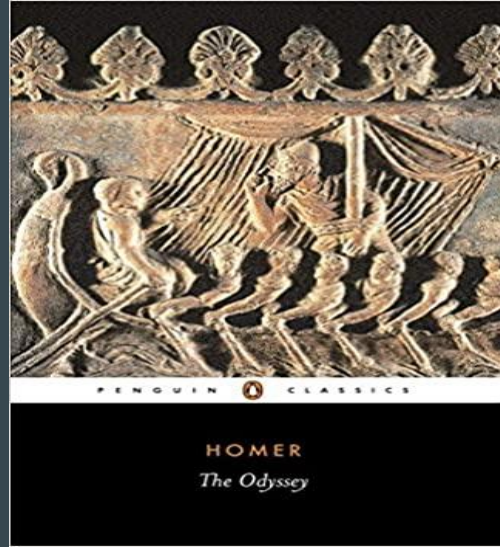
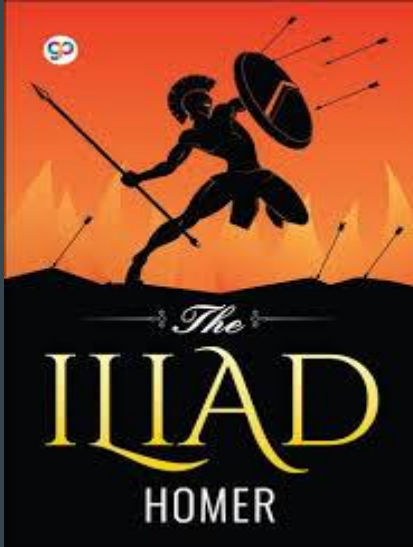
Ramayana Connection



Ramayana/HSSH connection

- Eldest son, who is the symbol of self-sacrifice (Mohnish Behl - Vivek)
 - it's an outsider who influences the mother to have the eldest son exiled. (property > *parivaar*.)
 - loyal middle brother who was not at home when the eldest brother is exiled. (Salman Khan - Prem)
 - *Bharat-milap (Vivek (MB) and Prem (SK))*
 - in *HSSH* Prem does not occupy the seat reserved for Vivek.
- *Hum Saath Saath Hain* also incorporated the sacrifice of Laxman's wife (a Slight variation is there)
 - Also, just like Ram Ji had the unwavering support of Hanuman, Vivek had the loyalty and friendship of Anwar.

Is there any Ramayana/Mahabharata for Europe/West?



Any Idea about this Remarkable Computer Scientist?

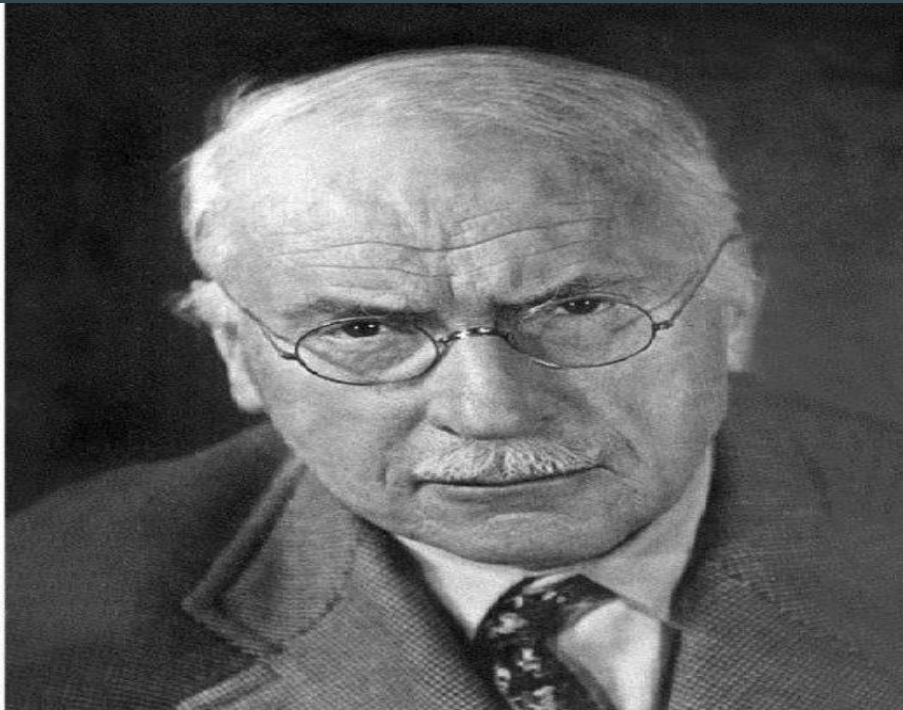
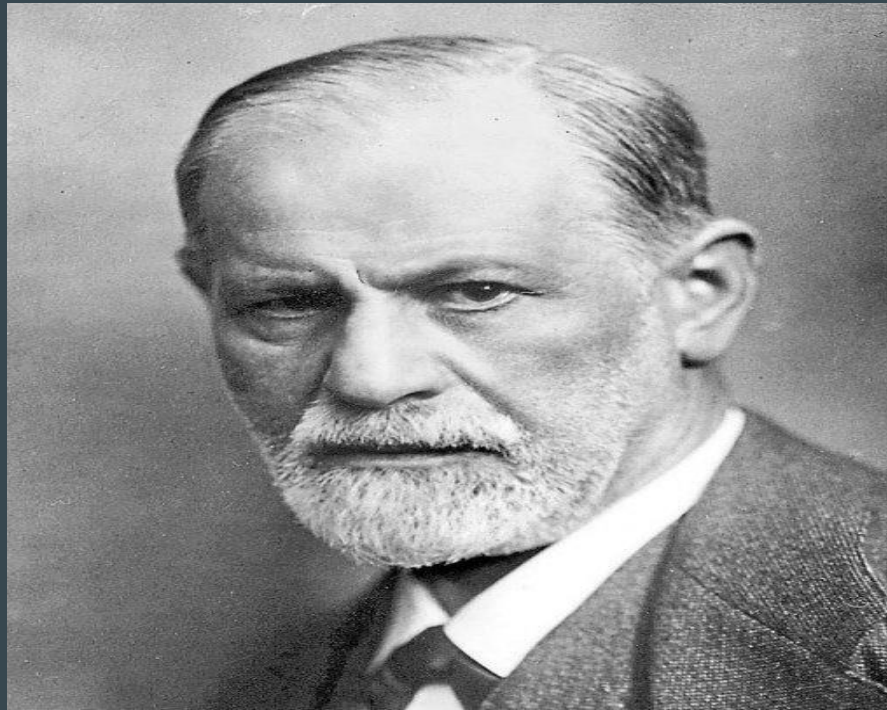


Alexander Stepanov – Civilizing the Barbarians



- Civilizing the Barbarians @A9.com
- He might be knowing more about India than us
- His talk series @
https://www.youtube.com/watch?v=c9BQnpaI58I&list=PLHxtyCq_WDLXzT_CgYPtjYab8fyd2PkjB

Who are They?



The Agenda

- Business Patterns which are ubiquitous
- Archetype is a term coined by the Legendary Swiss psychologist Carl Gustav Jung. (Jung was an expert in “Yogasutra” and Kundalini Yoga and it is widely believed that the idea has got its origins in Vedic term “Adhiroopa” and another term Synchronicity from the Vedic term “Ritu”!)
- •Archetype and Archetype Patterns

What is an Archetype?

- An Archetype is a primordial thing or Circumstances that recurs consistently and is thought to be a universal concept or situation
- It arises from the collective consciousness or common fund of human experiences
- Most archetypes show a trait called “Variability” to adapt to different cultural context.

Some Examples of Archetypes

- Some examples of Psychological archetypes are Hero , Tragically flawed hero , anti-hero , villain , girl with a perpetual misfortune, Victim, Cupid etc
- “There might be twenty odd archetypes in the world and eight/nine plots in the world. Permutations and Combinations of all these things is manifested as Stories” – A Remark by a friend when he heard about the Arlow/Nuestadt book.

What is Business?

- Business is supposed to be the third oldest profession in the world!
- Which is the first and the second?
- I will give the answer indirectly
- “You know, it has been said that politics is the second oldest profession and I've come to realize over the last few years, it bears a great similarity to the first.” (*Speech at Hillsdale College, Hillsdale, Michigan on November 10, 1977*)



Archetypes!..what it has to do with software ????

- Since Business is at least three millennia old , Archetypes like Party , customer , product , inventory , money do popup
- Such Archetypes can be called a Business Archetype.
- Enterprise modeling is all about modeling business scenarios.

The Idea of Business Archetype (from Arlow/Nuestadt book)

- A Business Archetype is a primordial thing that occurs consistently and universally in business domains and software systems.
- •The collaboration between business archetypes that occurs consistently and universally between business environment and software systems has to be modeled
- •How do we do that ?
- How do we model collaboration between Archetypes ?
- The ANSWER is PATTERNS (Plots)
- Patterns ??? –what the heck is it ?

What are Patterns?

- •“A Pattern describes a problem which occurs over and over again in the environment in question and then describes the core of the solution to the problem ,in such a way that one can use this solution a million times over ,without much thinking about it. “ (Adapted from Christopher Alexander’s definition –Christopher Alexander is an architect)

About Patterns and Pattern Movement

- Christopher Alexander and town building
- Erich Gamma et al... (GoF)
- Martin Fowler and analysis patterns
- Martin Fowler and Enterprise architecture patterns
- J2EE Core patterns from John crupi , deepak Alur
- Microsoft patterns and practices
- POSA catalog (five volumes!)
- Arlow/Nuestadt Business Archetype patterns....
- And many other domain specific patterns like data patterns , infrastructure patterns , health care , finance etc

Learn about Pattern Oriented Software from your CTO!

- GOF Pattern Catalog
- POEAA catalog
- EIP catalog
- J2EE Design Catalog
- DDD catalog
- POSA catalog
- Arlow/Nueststadt Catalog
- Covers Library development in Patterns, Concurrency Management, Reactive Programming and Object/Functional Programming (It is an Opinionated book!)

.NET Design Patterns

Knowing about design patterns enables developers to improve their code base, promoting code reuse and making their design more robust.

This book focuses on the practical aspects of programming in .NET. You will learn about some of the relevant design patterns (and their application) that are most widely used. We start with classic object-oriented programming (OOP) techniques, evaluate parallel programming and concurrency models, enhance implementations by mixing OOP and functional programming, and finally to the reactive programming model where functional programming and OOP are used in synergy to write better code.

Throughout this book, we'll show you how to deal with architecture/design techniques, GOF patterns, relevant patterns from other catalogs, functional programming, and reactive programming techniques.

After reading this book, you will be able to convincingly leverage these design patterns (factory pattern, builder pattern, prototype pattern, adapter pattern, facade pattern, decorator pattern, observer pattern and so on) for your programs. You will also be able to write fluid functional code in .NET that would leverage concurrency and parallelism!

Things you will learn:

- Put patterns and pattern catalogs into the right perspective
- Apply patterns for software development under C#/NET
- Use GoF and other patterns in real-life development scenarios
- Be able to enrich your design vocabulary and well articulate your design thoughts
- Leverage object/functional programming by mixing OOP and FP
- Understand the reactive programming model using Rx and RxJs
- Writing compositional code using C# LINQ constructs
- Be able to implement concurrent/parallel programming techniques using idioms under .NET
- Avoiding pitfalls when creating compositional, readable, and maintainable code using imperative, functional, and reactive code.

Packt
www.packtpub.com

\$ 39.99 US
£ 32.99 UK

Prices do not include local sales
Tax or VAT where applicable



What is a Business Archetype Pattern?

- A business archetype pattern is a collaboration between business archetypes that occurs consistently and universally in business environments and software systems.

Part 2

A short tour of Business Archetypes and Business Archetype patterns

The Party Archetype and related ones

- Business context
- Unique identifier , Registered identifier , Name , address-email address , geographic address , telephone # etc
- Party authentication and security violation
- Gender –male , female , not known, not specified
- Ethnicity etc

The Party Relationship archetype

- PartyRelationship can be quite abstract and complex
- Capturing relationship between parties is essential for CRM, organization units etc
- Party Role and Relationship
- Party Role type and Party relationship type
- Responsibilities
- Capabilities

Customer relationship management archetype patterns

- Customer (Party)
- CustomerService case
- Customer communication manager
- Customer Agent –how to represent the role of some one who acts on the behalf of the customer
- Action and outcome of the service case -Resolution

Quantity Archetype Pattern

- Which metric has to be used ? A metric is a standard of measurement
- SI system and MKS system
- Money and currency
- Unit and System of units
- Base Unit and derived units
- Man hour
- Quantity arithmetic –precision –rounding , comparison , floating point round off, standard conversion

Product Archetype Pattern

- Generalized model of representing products
- Product type and Product instance
- Product specification
- Product catalogue
- Packaging
- Rule driven packaging specification

Inventory Archetype Pattern

- How inventory archetype interacts with product
- How to represent inventory of items ?
- Inventory entry
- ProductInventoryentry –how to represent inventory information for a set of Product instances all of a same product type.
- ServiceInventoryentry –how to represent inventory information for a set of service instances all of same service type
- •Availability of product instance for sale
- •Reservation arrangement by which a product instance is reserved for a particular party.

Order Archetype

- Order archetype –order identifier
 - Purchase Order –sales order
 - Order Line
 - Party summary role in order
 - Charge Line and Tax On Line
 - Order events
 - Order status
- Order manager
 - Amend events
 - Amend Order Line events –
Change , Add , delete order line
 - DiscountEvent
 - DespatchEvent
 - OrderPayment and Payment
events
 - Other things to follow

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Business Rule Archetype Pattern

- How rules interact with the business process ?
- Business rules and the business system rules
- Business rules are captured in source code , data base triggers , explicit rules and implied rules
- Use a Rule Engine ! (Plugins , Scripting Languages or DSL)

Part 3 – Applications Survey

The Maven Archetypes for Building Software

- An archetype is defined as *an original pattern or model from which all other things of the same kind are made*. Archetype will help authors create Maven project templates for users, and provides users with the means to generate parameterized versions of those project templates.
- Using archetypes provides a great way to enable developers quickly in a way consistent with best practices employed by your project or organization. Within the Maven project we use archetypes to try and get our users up and running as quickly as possible by providing a sample project that demonstrates many of the features of Maven while introducing new users to the best practices employed by Maven. In a matter of seconds a new user can have a working Maven project to use as a jumping board for investigating more of the features in Maven. you may want to provide archetypes for EJBs, or WARs, or for your web services. Once these archetypes are created and deployed in your organization's repository they are available for use by all developers within your organization.

A Health Care Study – Ontology

- We created an Ontology to model Medical aid forms
- Identified Archetypes within the document and Questionnaire were formed on the Archetypes identified
- Later, the Answers were projected to 318 documents
- A Paper @ https://github.com/praseedpai/RuleEngineAndImplementation/blob/master/Ontology_On_Mobile_Paper.pdf

Questions

- If any!